1 Posterior Error Induced by Surrogate Model Eq(11)

Here we study how mis-specification of the surrogate observation model in Eq. (11) of the main text,

$$\varphi_{V,i}(V) = \mu_i(Z) + \xi_i, \qquad \xi_i \sim N(0, \eta^2),$$

can affect the posterior approximation error for the pointwise causal quantity $\gamma(w, z)$.

1.1 Set-up

Throughout, we fix a realized dataset $D = \{(V_i, Z_i, W_i, Y_i)\}_{i=1}^n$.. All statements below are conditional on this observed dataset, so the following analysis is finite-sample rather than asymptotic.

Causal Effect of Interest Using the notation of the main paper, recall that the causal effects of interest can be expressed as

$$\gamma(w, z) = f\big(\psi_W(w) \otimes \mu(z)\big),$$

where

$$\mu(z) := \mathbb{E}[\psi_V(V) \mid Z = z] \in H_V$$

is the conditional mean embedding of V given Z. By the Mercer expansion of $\psi_V(v) = k_V(\cdot, v)$, this admits the spectral representation

$$\mu(z) = \sum_{i \in I} \lambda_{V,i} \mu_i(z) \varphi_{V,i}(\cdot), \qquad \mu_i(z) := \mathbb{E}[\varphi_{V,i}(V) \mid Z = z].$$

Thus, for each fixed (w, z), the scalar quantity $\gamma(w, z)$ is determined by the latent pair (f, μ). We therefore define

$$T_{w,z}(f, \mu) := f\big(\psi_W(w) \otimes \mu(z)\big),$$

so that

$$\gamma(w, z) = T_{w,z}(f, \mu).$$

Probabilistic model. Let Π denote the prior distribution of the latent pair (f, μ) induced by the GP priors in the main text. Let $\hat{L}$ denote the likelihood induced by the surrogate observation model in Eq. (11), and let L denote a *well-specified*[1] benchmark likelihood on the latent pair (f, μ). The purpose of the following derivation is to quantify the posterior mis-specification error induced by replacing the well-specified likelihood L with the surrogate likelihood $\hat{L}$.

Define the corresponding posteriors

$$P(df, d\mu) := \frac{L(f, \mu)\Pi(df, d\mu)}{Z}, \qquad \hat{P}(df, d\mu) := \frac{\hat{L}(f, \mu)\Pi(df, d\mu)}{\hat{Z}}.$$

with normalizing constants $Z := \int L(f, \mu)\, \Pi(df, d\mu)$, $\hat{Z} := \int \hat{L}(f, \mu)\, \Pi(df, d\mu)$, assumed finite and strictly positive. Now, define the likelihood ratio,

$$R(f, \mu) := \frac{\hat{L}(f, \mu)}{L(f, \mu)}.$$

and note that we can express the normalizing constant ratio as follows

$$\frac{\hat{Z}}{Z} = \int \frac{\hat{L}(f, \mu)}{L(f, \mu)}\, P(df, d\mu) = \int R(f, \mu)\, P(df, d\mu) = \mathbb{E}_P[R].$$

[1] By this we mean that there exists a latent pair $(f^\star, \mu^\star)$ such that the true conditional distribution of the observed data is exactly $L(\cdot \mid f^\star, \mu^\star)$. In contrast, $\hat{L}$ is the surrogate likelihood induced by Eq. (11), and need not coincide with the true conditional distribution.

Hence $\hat{P} \ll P$ and the Radon-Nikodym derivative can be expressed in terms of R.

$$\frac{d\hat{P}}{dP}(f,\mu) = \frac{R(f,\mu)}{\mathbb{E}_P[R]}.$$

The above identities let us bound several divergences/metrics of interest in terms of functionals of R,

$$\mathrm{KL}(\hat{P}\|P) = \int \log\left(\frac{d\hat{P}}{dP}\right) d\hat{P} = \mathbb{E}_{\hat{P}}[\log R] - \log \mathbb{E}_P[R].$$

$$\mathrm{TV}(P,\hat{P}) = \frac{1}{2}\int \left|\frac{d\hat{P}}{dP} - 1\right| dP = \frac{1}{2}\,\mathbb{E}_P\left[\left|\frac{R}{\mathbb{E}_P[R]} - 1\right|\right].$$

$$\chi^2(\hat{P}\|P) := \int \left(\frac{d\hat{P}}{dP} - 1\right)^2 dP = \mathbb{E}_P\left[\left(\frac{R}{\mathbb{E}_P[R]} - 1\right)^2\right].$$

In what follows, we will use these identities to bound how likelihood differences propagate to posterior differences.

1.2 Joint Posterior Bounds under log-likelihood-ratio control

Assume the log-likelihood ratio between the well-specified and surrogate likelihood can be bounded as follows for some $\varepsilon > 0$

$$|\log R(f,\mu)| \le \varepsilon \qquad \text{for all } (f,\mu) \in \mathcal{H}.$$

This bound can be used to control the posterior divergences above. In particular, since $e^{-\varepsilon} \le R(f,\mu) \le e^{\varepsilon}$ for all (f,μ), we have

$$e^{-\varepsilon} \le \mathbb{E}_P[R] \le e^{\varepsilon}.$$

Using the exact KL identity,

$$\mathrm{KL}(\hat{P}\|P) = \mathbb{E}_{\hat{P}}[\log R] - \log \mathbb{E}_P[R] \le \varepsilon + \varepsilon = 2\varepsilon.$$

Also,

$$\chi^2(\hat{P}\|P) = \int \left(\frac{d\hat{P}}{dP} - 1\right)^2 dP = \mathbb{E}_P\left[\left(\frac{R}{\mathbb{E}_P[R]} - 1\right)^2\right] \le (e^{2\varepsilon} - 1)^2.$$

Finally, by Pinsker's inequality and the bound $\mathrm{TV}(P,\hat{P}) \le \sqrt{\chi^2(\hat{P}\|P)}$, the TV is bounded by

$$\mathrm{TV}(P,\hat{P}) \le \min\left\{\sqrt{\frac{1}{2}\,\mathrm{KL}(\hat{P}\|P)}, \sqrt{\chi^2(\hat{P}\|P)}\right\} \le \min\left\{\sqrt{\varepsilon},\, e^{2\varepsilon} - 1\right\}.$$

1.3 Propagation to the posterior of causal effect $\gamma(w,z)$

The above control on the posterior over latent processes propagates automatically to the posterior of the causal effect $\gamma(w,z)$. In particular, let

$$P_{w,z} := P \circ T_{w,z}^{-1}, \qquad \hat{P}_{w,z} := \hat{P} \circ T_{w,z}^{-1}$$

denote the induced posteriors of the scalar quantity $\gamma_{w,z} = \gamma(w,z)$ under P and $\hat{P}$.

For total variation, we immediately get the contraction bound

$$\mathrm{TV}(P_{w,z},\hat{P}_{w,z}) = \sup_{B\in\mathcal{B}(\mathbb{R})} |P(T_{w,z}^{-1}(B)) - \hat{P}(T_{w,z}^{-1}(B))| \le \sup_{A\in\mathcal{F}} |P(A) - \hat{P}(A)| = \mathrm{TV}(P,\hat{P}).$$

Since both KL and χ^2 are f-divergences, the data-processing inequality for the measurable map $T_{w,z}$ also gives

$$\mathrm{KL}(\hat{P}_{w,z}\|P_{w,z}) \le \mathrm{KL}(\hat{P}\|P), \qquad \chi^2(\hat{P}_{w,z}\|P_{w,z}) \le \chi^2(\hat{P}\|P).$$

Combining this with the previous bounds yields

$$\mathrm{KL}(\hat{P}_{w,z}\|P_{w,z}) \le 2\varepsilon, \qquad \chi^2(\hat{P}_{w,z}\|P_{w,z}) \le (e^{2\varepsilon} - 1)^2, \qquad \mathrm{TV}(P_{w,z},\hat{P}_{w,z}) \le \min\left\{\sqrt{\varepsilon},\, e^{2\varepsilon} - 1\right\}.$$

Remark. Let $\tilde{P}_{w,z}$ denote the Gaussian moment-matching approximation to the surrogate posterior $\hat{P}_{w,z}$ used in the main text to construct credible intervals from its posterior moments. Then, by the triangle inequality for total variation,

$$\mathrm{TV}(P_{w,z}, \tilde{P}_{w,z}) \le \mathrm{TV}(P_{w,z}, \hat{P}_{w,z}) + \mathrm{TV}(\hat{P}_{w,z}, \tilde{P}_{w,z}) \le \min\Big\{\sqrt{\varepsilon},\, e^{2\varepsilon} - 1\Big\} + \mathrm{TV}(\hat{P}_{w,z}, \tilde{P}_{w,z}).$$

Thus, under the stated log-likelihood-ratio assumption, the additional total posterior error induced by replacing the benchmark likelihood with the surrogate likelihood is at most the bound derived above; any remaining error is due to the Gaussian projection step.

1.4 Effect on Posterior Moments

Since the Gaussian approximation used in the main text is matched to the posterior mean and covariance of the surrogate posterior, it is natural to study how the above likelihood error can affect these moments.

Bound based on χ^2. For brevity, define

$$\delta_\varepsilon := e^{2\varepsilon} - 1.$$

Now suppose $X \in L^4(P_{w,z})$. Then, since $\int \left(\frac{d\hat{P}_{w,z}}{dP_{w,z}} - 1\right) dP_{w,z} = 0$, for any measurable $h \in L^2(P_{w,z})$

$$\begin{aligned}
\mathbb{E}_{\hat{P}_{w,z}}[h(X)] - \mathbb{E}_{P_{w,z}}[h(X)] &= \int h(x) \left(\frac{d\hat{P}_{w,z}}{dP_{w,z}}(x) - 1\right) P_{w,z}(dx) \\
&= \int \big(h(x) - \mathbb{E}_{P_{w,z}}[h(X)]\big) \left(\frac{d\hat{P}_{w,z}}{dP_{w,z}}(x) - 1\right) P_{w,z}(dx),
\end{aligned}$$

Now, by Cauchy–Schwarz,

$$\Big|\mathbb{E}_{\hat{P}_{w,z}}[h(X)] - \mathbb{E}_{P_{w,z}}[h(X)]\Big| \le \sqrt{\chi^2(\hat{P}_{w,z} \| P_{w,z})}\, \sqrt{\mathrm{Var}_{P_{w,z}}(h(X))}.$$

Using the bound on $\chi^2(\hat{P}_{w,z} \| P_{w,z})$, this gives

$$\Big|\mathbb{E}_{\hat{P}_{w,z}}[h(X)] - \mathbb{E}_{P_{w,z}}[h(X)]\Big| \le \delta_\varepsilon \sqrt{\mathrm{Var}_{P_{w,z}}(h(X))}.$$

Applying this with $h(x) = x$ and then $h(x) = x^2$ yields the first and second moment bounds

$$\Big|\mathbb{E}_{\hat{P}_{w,z}}[X] - \mathbb{E}_{P_{w,z}}[X]\Big| \le \delta_\varepsilon \sqrt{\mathrm{Var}_{P_{w,z}}(X)}.$$

$$\Big|\mathbb{E}_{\hat{P}_{w,z}}[X^2] - \mathbb{E}_{P_{w,z}}[X^2]\Big| \le \delta_\varepsilon \sqrt{\mathrm{Var}_{P_{w,z}}(X^2)}.$$

Finally, writing

$$\Delta_1 := \mathbb{E}_{\hat{P}_{w,z}}[X] - \mathbb{E}_{P_{w,z}}[X], \qquad \Delta_2 := \mathbb{E}_{\hat{P}_{w,z}}[X^2] - \mathbb{E}_{P_{w,z}}[X^2],$$

we have

$$\begin{aligned}
\Big|\mathrm{Var}_{\hat{P}_{w,z}}(X) - \mathrm{Var}_{P_{w,z}}(X)\Big| &= \Big|\Delta_2 - \Big((\mathbb{E}_{\hat{P}_{w,z}}[X])^2 - (\mathbb{E}_{P_{w,z}}[X])^2\Big)\Big| \\
&\le |\Delta_2| + |\Delta_1| \Big(|\mathbb{E}_{\hat{P}_{w,z}}[X]| + |\mathbb{E}_{P_{w,z}}[X]|\Big) \\
&\le |\Delta_2| + |\Delta_1| \Big(2|\mathbb{E}_{P_{w,z}}[X]| + |\Delta_1|\Big),
\end{aligned}$$

since $|\mathbb{E}_{\hat{P}_{w,z}}[X]| \le |\mathbb{E}_{P_{w,z}}[X]| + |\Delta_1|$. Substituting the bounds on Δ_1 and Δ_2 gives

$$\Big|\mathrm{Var}_{\hat{P}_{w,z}}(X) - \mathrm{Var}_{P_{w,z}}(X)\Big| \le \delta_\varepsilon \sqrt{\mathrm{Var}_{P_{w,z}}(X^2)} + 2|\mathbb{E}_{P_{w,z}}[X]|\, \delta_\varepsilon \sqrt{\mathrm{Var}_{P_{w,z}}(X)} + \delta_\varepsilon^2\, \mathrm{Var}_{P_{w,z}}(X).$$

Bound based on TV. Let

$$X := \gamma(w, z), \qquad P_{w,z} := P \circ T_{w,z}^{-1}, \qquad \hat{P}_{w,z} := \hat{P} \circ T_{w,z}^{-1}.$$

Set

$$\nu := \hat{P}_{w,z} - P_{w,z}.$$

Then ν is a finite signed measure with total variation measure $|\nu|$, and

$$|\nu|(\mathbb{R}) = 2\,\mathrm{TV}(\hat{P}_{w,z}, P_{w,z}).$$

Moreover, since $\nu^+ \le \hat{P}_{w,z}$ and $\nu^- \le P_{w,z}$, we have

$$|\nu| = \nu^+ + \nu^- \le \hat{P}_{w,z} + P_{w,z}.$$

Now let $p > 1$ and $q := p/(p-1)$. For any measurable h with

$$h \in L^p(P_{w,z}) \cap L^p(\hat{P}_{w,z}),$$

Hölder's inequality with respect to $|\nu|$ gives

$$\begin{aligned}\left|\mathbb{E}_{\hat{P}_{w,z}}[h(X)] - \mathbb{E}_{P_{w,z}}[h(X)]\right| &= \left|\int h\, d\nu\right| \le \int |h|\, d|\nu| \\ &\le \left(\int |h|^p\, d|\nu|\right)^{1/p} |\nu|(\mathbb{R})^{1/q}.\end{aligned}$$

Using $|\nu| \le \hat{P}_{w,z} + P_{w,z}$ and $|\nu|(\mathbb{R}) = 2\,\mathrm{TV}(\hat{P}_{w,z}, P_{w,z})$, this yields

$$\left|\mathbb{E}_{\hat{P}_{w,z}}[h(X)] - \mathbb{E}_{P_{w,z}}[h(X)]\right| \le \left(\mathbb{E}_{P_{w,z}}|h(X)|^p + \mathbb{E}_{\hat{P}_{w,z}}|h(X)|^p\right)^{1/p} \left(2\,\mathrm{TV}(\hat{P}_{w,z}, P_{w,z})\right)^{1-1/p}.$$

In particular, if

$$\mathbb{E}_{P_{w,z}}[|h(X)|^p] \le M_p, \qquad \mathbb{E}_{\hat{P}_{w,z}}[|h(X)|^p] \le M_p,$$

then

$$\left|\mathbb{E}_{\hat{P}_{w,z}}[h(X)] - \mathbb{E}_{P_{w,z}}[h(X)]\right| \le 2\, M_p^{1/p}\, \mathrm{TV}(\hat{P}_{w,z}, P_{w,z})^{\,1-1/p}.$$

Corollaries for the posterior mean and variance. If

$$\mathbb{E}_{P_{w,z}}[X^2],\ \mathbb{E}_{\hat{P}_{w,z}}[X^2] \le M_2,$$

then taking $h(x) = x$ and $p = 2$ gives

$$\left|\mathbb{E}_{\hat{P}_{w,z}}[X] - \mathbb{E}_{P_{w,z}}[X]\right| \le 2\, M_2^{1/2}\, \mathrm{TV}(\hat{P}_{w,z}, P_{w,z})^{1/2}.$$

If

$$\mathbb{E}_{P_{w,z}}[X^4],\ \mathbb{E}_{\hat{P}_{w,z}}[X^4] \le M_4,$$

then taking $h(x) = x^2$ and $p = 2$ gives

$$\left|\mathbb{E}_{\hat{P}_{w,z}}[X^2] - \mathbb{E}_{P_{w,z}}[X^2]\right| \le 2\, M_4^{1/2}\, \mathrm{TV}(\hat{P}_{w,z}, P_{w,z})^{1/2}.$$

Hence

$$\begin{aligned}\left|\mathrm{Var}_{\hat{P}_{w,z}}(X) - \mathrm{Var}_{P_{w,z}}(X)\right| &\le \left|\mathbb{E}_{\hat{P}_{w,z}}[X^2] - \mathbb{E}_{P_{w,z}}[X^2]\right| + \left|(\mathbb{E}_{\hat{P}_{w,z}}[X])^2 - (\mathbb{E}_{P_{w,z}}[X])^2\right| \\ &\le 2\, M_4^{1/2}\, \mathrm{TV}(\hat{P}_{w,z}, P_{w,z})^{1/2} + 4\, M_2\, \mathrm{TV}(\hat{P}_{w,z}, P_{w,z})^{1/2}.\end{aligned}$$